

Topic: OOP



Bachelor of Engineering (Software Engineering)

Semester 2

Session (2025-2029)

Assignment no# 1

Submitted By:

Eman Fatima (2521252261)

Submitted to:

Dr Mohsin Nazir

Department of Software Engineering

Lahore College for Women University, Lahore

HOTEL MAMGEMENT SYSTEM WITH OUTPUT

```
#include<iostream>

#include<string.h>


#define max 100

using name space std;


//Class Customer
class Customer
{
public:
char name[100];
char address[100];
char phone[12];
char from_date[20];
char to_date[20];
float payment_advance;
int booking_id;
};


class Room
{
```

```
public:  
char type;  
char stype;  
char ac;  
int roomNumber;  
int rent;  
int status;
```

```
class Customer cust;  
class Room addRoom(int);  
void searchRoom(int);  
void deleteRoom(int);  
void displayRoom(Room);  
};
```

```
//Global Declarations  
class Room rooms[max];  
int count=0;
```

```
Room Room::addRoom(int rno)  
{  
class Room;  
room.roomNumber=rno;
```

```
cout<<"\nType AC/Non-AC (A/N) : ";
cin>>room.ac;
cout<<"\nType Comfort (S/N) : ";
cin>>room.type;
cout<<"\nType Size (B/S) : ";
cin>>room.stype;
cout<<"\nDaily Rent : ";
cin>>room.rent;
room.status=0;

cout<<"\n Room Added Successfully!";
return room;
}
```

```
void Room::searchRoom(int rno)
{
int i,found=0;
for(i=0;i<count;i++)
{
if(rooms[i].roomNumber==rno)
{
found=1;
break;
}
```

```
}  
  
}  
  
if(found==1)  
{  
    cout<<"Room Details\n";  
    if(rooms[i].status==1)  
    {  
        cout<<"\nRoom is Reserved";  
    }  
    else  
    {  
        cout<<"\nRoom is available";  
    }  
    displayRoom(rooms[i]);  
}  
else  
{  
    cout<<"\nRoom not found";  
}  
}  
  
void Room::displayRoom(Room tempRoom)  
{  
    cout<<"\nRoom Number: \t"<<tempRoom.roomNumber;
```

```
cout<<"\nType AC/Non-AC (A/N) "<<tempRoom.ac;
cout<<"\nType Comfort (S/N) "<<tempRoom.type;
cout<<"\nType Size (B/S) "<<tempRoom.stype;
cout<<"\nRent: "<<tempRoom.rent;
}
```

```
//hotel management class
```

```
class HotelMgnt:protected Room
```

```
{
```

```
public:
```

```
void checkIn();
```

```
void getAvailRoom();
```

```
void searchCustomer(char *);
```

```
void checkOut(int);
```

```
void guestSummaryReport();
```

```
};
```

```
void HotelMgnt::guestSummaryReport(){
```

```
if(count==0){
```

```
    cout<<"\n No Guest in Hotel !!";
```

```
}
```

```
for(int i=0;i<count;i++)
```

```

{
if(rooms[i].status==1)
{
cout<<"\n Customer First Name : "<<rooms[i].cust.name;
cout<<"\n Room Number : "<<rooms[i].roomNumber;
cout<<"\n Address (only city) : "<<rooms[i].cust.address;
cout<<"\n Phone : "<<rooms[i].cust.phone;
cout<<"\n-----";
}

}

}

```

//hotel management reservation of room

```
void HotelMgnt::checkIn()
```

```

{
int i,found=0,rno;

```

```
class Room;
```

```
cout<<"\nEnter Room number : ";
```

```
cin>>rno;
```

```
for(i=0;i<count;i++)
```

```
{
```

```
if(rooms[i].roomNumber==rno)
{
found=1;
break;
}
}
if(found==1)
{
if(rooms[i].status==1)
{
cout<<"\nRoom is already Booked";
return;
}
```

```
cout<<"\nEnter booking id: ";
cin>>rooms[i].cust.booking_id;
```

```
cout<<"\nEnter Customer Name (First Name): ";
cin>>rooms[i].cust.name;
```

```
cout<<"\nEnter Address (only city): ";
cin>>rooms[i].cust.address;
```

```
cout<<"\nEnter Phone: ";
```

```
cin>>rooms[i].cust.phone;
```

```
cout<<"\nEnter From Date: ";
```

```
cin>>rooms[i].cust.from_date;
```

```
cout<<"\nEnter to Date: ";
```

```
cin>>rooms[i].cust.to_date;
```

```
cout<<"\nEnter Advance Payment: ";
```

```
cin>>rooms[i].cust.payment_advance;
```

```
rooms[i].status=1;
```

```
cout<<"\n Customer Checked-in Successfully..";
```

```
}
```

```
}
```

```
//hotel management shows available rooms
```

```
void HotelMgnt::getAvailRoom()
```

```
{
```

```
int i,found=0;
```

```
for(i=0;i<count;i++)
```

```
{  
if(rooms[i].status==0)  
{  
displayRoom(rooms[i]);  
cout<<"\n\nPress enter for next room";  
found=1;  
}  
}  
if(found==0)  
{  
cout<<"\nAll rooms are reserved";  
}  
}
```

//hotel management shows all persons that have booked room

```
void HotelMgnt::searchCustomer(char *pname)  
{  
int i,found=0;  
for(i=0;i<count;i++)  
{  
if(rooms[i].status==1 && strcmp(rooms[i].cust.name,pname)==0)  
{  
cout<<"\nCustomer Name: "<<rooms[i].cust.name;
```

```
cout<<"\nRoom Number: "<<rooms[i].roomNumber;
```

```
cout<<"\n\nPress enter for next record\n";
```

```
found=1;
```

```
}
```

```
}
```

```
if(found==0)
```

```
{
```

```
cout<<"\nPerson not found.\n";
```

```
}
```

```
}
```

```
//hotel managemt generates the bill of the expenses
```

```
void HotelMgnt::checkOut(int roomNum)
```

```
{
```

```
int i,found=0,days,rno;
```

```
float billAmount=0;
```

```
for(i=0;i<count;i++)
```

```
{
```

```
if(rooms[i].status==1 && rooms[i].roomNumber==roomNum)
```

```
{
```

```
//rno = rooms[i].roomNumber;
```

```
found=1;
```

```
//getch();  
break;  
}  
}  
if(found==1)  
{  
cout<<"\nEnter Number of Days:\t";  
cin>>days;  
billAmount=days * rooms[i].rent;  
  
cout<<"\n\t##### CheckOut Details #####\n";  
cout<<"\nCustomer Name : "<<rooms[i].cust.name;  
cout<<"\nRoom Number : "<<rooms[i].roomNumber;  
cout<<"\nAddress : "<<rooms[i].cust.address;  
cout<<"\nPhone : "<<rooms[i].cust.phone;  
cout<<"\nTotal Amount Due : "<<billAmount<<" /";  
cout<<"\nAdvance Paid: "<<rooms[i].cust.payment_advance<<" /";  
cout<<"\n*** Total Payable: "<<billAmount-rooms[i].cust.payment_advance<<" / only";  
  
rooms[i].status=0;  
}  
}
```

```
//managing rooms (adding and searching available rooms)
```

```
void manageRooms()
```

```
{
```

```
class Room;
```

```
int opt,rno,i,flag=0;
```

```
char ch;
```

```
do
```

```
{
```

```
cout<<"\n### Manage Rooms ###";
```

```
cout<<"\n1. Add Room";
```

```
cout<<"\n2. Search Room";
```

```
cout<<"\n3. Back to Main Menu";
```

```
cout<<"\n\nEnter Option: ";
```

```
cin>>opt;
```

```
//switch statement
```

```
switch(opt)
```

```
{
```

```
case 1:
```

```
cout<<"\nEnter Room Number: ";
```

```
cin>>rno;
```

```
i=0;
```

```
for(i=0;i<count;i++)
```

```
{
if(rooms[i].roomNumber==rno)
{
flag=1;
}
}
if(flag==1)
{
cout<<"\nRoom Number is Present.\nPlease enter unique Number";
flag=0;
}
else
{
rooms[count]=room.addRoom(rno);
count++;
}
break;
case 2:
cout<<"\nEnter room number: ";
cin>>rno;
room.searchRoom(rno);
break;
case 3:
//nothing to do
```

```
break;

default:

cout<<"\nPlease Enter correct option";

break;

}

}while(opt!=3);

}

using namespace std;

int main()

{

class HotelMgnt hm;

int i,j,opt,rno;

char ch;

char pname[100];


do

{

cout<<"##### Hotel Management #####\n";

cout<<"\n1. Manage Rooms";

cout<<"\n2. Check-In Room";

cout<<"\n3. Available Rooms";

cout<<"\n4. Search Customer";

cout<<"\n5. Check-Out Room";
```

```
cout<<"\n6. Guest Summary Report";
cout<<"\n7. Exit";
cout<<"\n\nEnter Option: ";
cin>>opt;
switch(opt)
{
case 1:
manageRooms();
break;
case 2:
if(count==0)
{
cout<<"\nRooms data is not available.\nPlease add the rooms first.";
}
else
hm.checkIn();
break;
case 3:
if(count==0)
{
cout<<"\nRooms data is not available.\nPlease add the rooms first.";
}
else
hm.getAvailRoom();
```

```
break;

case 4:
if(count==0)
{
cout<<"\nRooms are not available.\nPlease add the rooms first.";
}
else
{
cout<<"Enter Customer Name: ";
cin>>pname;
hm.searchCustomer(pname);
}
break;

case 5:
if(count==0)
{
cout<<"\nRooms are not available.\nPlease add the rooms first.";
}
else
{
cout<<"Enter Room Number : ";
cin>>rno;
hm.checkOut(rno);
}
```

```
break;

case 6:
    hm.guestSummaryReport();
    break;

case 7:
    cout<<"\nTHANK YOU! FOR USING SOFTWARE\n";
    break;

default:
    cout<<"\nPlease Enter correct option";
    break;
}

}while(opt!=7);

}
```

OUTPUT



C:\Users\PMLS\Pictures\X.exe



Hotel Management

1. Manage Rooms
2. Check-In Room
3. Available Rooms
4. Search Customer
5. Check-Out Room
6. Guest Summary Report
7. Exit

Enter Option:



C:\Users\PMLS\Pictures\X.exe



Hotel Management

1. Manage Rooms
2. Check-In Room
3. Available Rooms
4. Search Customer
5. Check-Out Room
6. Guest Summary Report
7. Exit

Enter Option: 1

Manage Rooms

1. Add Room
2. Search Room
3. Back to Main Menu

Enter Option: 1

Enter Room Number: 1

Type AC/Non-AC (A/N) : AC

Type Comfort (S/N) :

Type Size (B/S) : S